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# HVDC Load Flow

DC power flow, current and losses for symmetrical monopole and bipole configurations



## Revision history

| REV. | DATE         | AUTHOR           | DESCRIPTION      |
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# 1 Scope and conventions

## 1.1 Purpose

This document records the calculation process behind the **Load Flow** tab of the HVDC calculation workbook. The tab answers one question for a point-to-point HVDC link: *given the AC power to be delivered at the receiving end, what DC current flows, what voltage arrives at the receiving converter, and what does the link cost in losses?* Two conductor arrangements are covered — the **symmetrical monopole** and the **bipole with a dedicated metallic return (DMR)** — together with the reduced-capacity operating case of a bipole that has lost one pole.

The intent is a screening-level tool: closed-form, one operating point at a time, transparent enough to check by hand, and consistent with the loss calculation already carried in the **Losses** tab of the same workbook. It is not a substitute for a network study; Section 7 lists what is deliberately left out.

## 1.2 Direction and reference conventions

Power flows from the **rectifier** (sending station) to the **inverter** (receiving station). Three conventions fix the problem and are used throughout:

- (1) **Power is defined at the receiving end**, at the AC terminals of the inverter station. This is the quantity a transmission service agreement or an interconnection request is written against, so it is the natural input; everything upstream — current, line losses, sending-end power — is then a result rather than an assumption.
- (2) **DC voltage is regulated at the rectifier DC terminals**. The nominal pole-to-ground voltage  $V_d$  holds at the sending station and the receiving-end voltage comes out lower by the resistive drop of the line.
- (3) **Converter station losses are a fraction of station throughput**, applied per station, rather than a fixed number of megawatts.

Conventions (1) and (2) sit at opposite ends of the line, which is what makes the current an implicit rather than an explicit quantity: the sending-end power depends on the line loss, the line loss depends on the current, and the current depends on the sending-end power. Section 4 closes that loop in one step instead of iterating.

## 1.3 Notation

### 1.4 Units

The calculation is carried in the engineering units the workbook uses, and the unit set is self-consistent without conversion factors provided currents are handled in kiloamperes:

$$[\text{kV}] \times [\text{kA}] = \text{MW}, \quad [\Omega] \times [\text{kA}]^2 = \text{MW}.$$

Currents are *reported* in amperes, which is why the spreadsheet formulas carry factors of 1000 and 1/1000 around the current cell. Every equation in this document uses  $I_d$  in kiloamperes.

The one genuine conversion concerns the conductor data. Conductor tables give DC resistance per 1000 ft, and that is the form entered in the workbook, so the factor

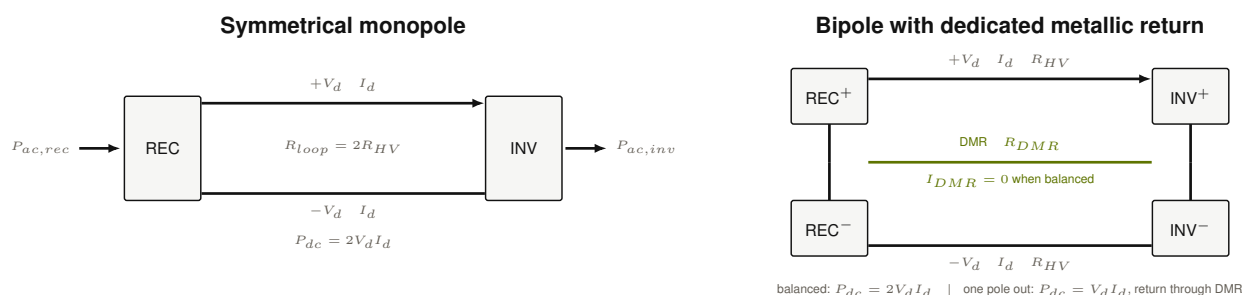
$$\frac{5280 \text{ ft/mi}}{1000 \text{ ft}} = 5.28$$

converts specific resistance to  $\Omega/\text{mi}$  before multiplying by the line length in miles.

Note: the corresponding input row on the **Losses** tab is labelled  $\Omega/\text{mi}$ , but the 5.28 factor in its loss formula shows the value entered there is in fact  $\Omega/1000 \text{ ft}$  (Bluebird ACSR,  $\approx 0.00801 \Omega/1000 \text{ ft}$ ). The Load Flow tab labels the row  $\Omega/1000 \text{ ft}$ ; the label on the Losses tab should be corrected to match.

| Symbol          | Unit             | Quantity                                                             |
|-----------------|------------------|----------------------------------------------------------------------|
| $P_{ac,inv}$    | MW               | AC power at the receiving end (inverter AC terminals) — <i>input</i> |
| $P_{ac,rec}$    | MW               | AC power drawn at the sending end (rectifier AC terminals)           |
| $P_{dc,inv}$    | MW               | DC power at the inverter DC terminals                                |
| $P_{dc,rec}$    | MW               | DC power at the rectifier DC terminals                               |
| $V_d$           | kV               | DC voltage pole-to-ground at the rectifier — <i>input</i>            |
| $V_{d,inv}$     | kV               | DC voltage pole-to-ground at the inverter                            |
| $I_d$           | kA               | DC pole current, operating value                                     |
| $I_{rated}$     | kA               | DC pole current including design margin                              |
| $L$             | mi               | DC line length                                                       |
| $r'_{HV}$       | $\Omega/1000$ ft | Specific resistance of one HV subconductor                           |
| $n_{HV}$        | —                | Subconductors per pole (bundle count)                                |
| $R_{HV}$        | $\Omega$         | Resistance of one pole conductor over the whole length               |
| $r'_{DMR}$      | $\Omega/1000$ ft | Specific resistance of one DMR subconductor                          |
| $n_{DMR}$       | —                | Subconductors in the DMR bundle                                      |
| $R_{DMR}$       | $\Omega$         | Resistance of the DMR over the whole length                          |
| $R_{loop}$      | $\Omega$         | Resistance of the complete current path (out and return)             |
| $p_{rec}$       | —                | Rectifier station loss fraction of throughput                        |
| $p_{inv}$       | —                | Inverter station loss fraction of throughput                         |
| $m$             | —                | Margin applied to the DC current for rating purposes                 |
| $P_{loss,line}$ | MW               | Total resistive loss of the DC line                                  |
| $\eta$          | —                | Overall AC-to-AC efficiency of the link                              |

**Table 1:** Notation used in the load flow calculation.



**Figure 1:** The two conductor arrangements. In both normal cases the current path is a two-conductor loop of resistance  $2R_{HV}$ ; only when a bipole pole is lost does the return move to the DMR and the loop resistance become  $R_{HV} + R_{DMR}$ .

## 2 Configurations covered

Figure 1 shows the two arrangements the calculation covers, and the current path each of them presents to the flow.

### 2.1 Symmetrical monopole

The symmetrical monopole carries the full power on two conductors held at  $+V_d$  and  $-V_d$  against ground, with no third conductor and no ground path in service. The DC circuit is a single loop: current leaves the rectifier on the positive conductor and returns on the negative one. The voltage seen by the converter is therefore the pole-to-pole voltage  $2V_d$ , and

$$P_{dc} = 2V_d I_d.$$

Because there is no return path other than the second conductor, the loss of either conductor is the loss of the whole link — there is no partial operating mode, which is the reason a bipole is preferred where availability matters.

### 2.2 Bipole with dedicated metallic return

The bipole runs two independently controlled poles at  $\pm V_d$  plus a third, lower-rated conductor: the dedicated metallic return. In normal **balanced** operation the two pole currents are equal and opposite, the DMR carries no current, and the circuit is electrically identical to the symmetrical monopole:

$$P_{dc} = 2V_d I_d, \quad I_{DMR} = 0.$$

The DMR earns its place when the bipole becomes unbalanced. With one pole out of service the remaining pole keeps operating and its current returns through the DMR instead of the lost conductor, so the link continues at reduced capacity. In that condition the circuit is a single-pole loop:

$$P_{dc} = V_d I_d,$$

and the return path resistance is that of the DMR rather than that of a second HV bundle — which is why the DMR conductor is a separate input in the calculation, as it is generally not the same conductor as the poles.

### 2.3 Reduced-capacity case as calculated

The one-pole-out case can be posed two ways: hold the power and double the pole current, or hold the current and accept roughly half the power. The workbook takes the second — the outage case runs at the *same pole current* as the balanced bipole, which keeps the case within the conductor and valve ratings already established for normal operation and needs no separate rating input. Power is then a result of that current, and comes out slightly below half of the bipolar figure because the return path through the DMR is more resistive than the parallel HV bundle it replaces.

| Configuration        | $P_{dc}$  | $R_{loop}$         | Current path                                                   |
|----------------------|-----------|--------------------|----------------------------------------------------------------|
| Symmetrical monopole | $2V_dI_d$ | $2R_{HV}$          | Out on the positive conductor, back on the negative conductor. |
| Bipole, balanced     | $2V_dI_d$ | $2R_{HV}$          | Out on the positive pole, back on the negative pole; DMR idle. |
| Bipole, one pole out | $V_dI_d$  | $R_{HV} + R_{DMR}$ | Out on the healthy pole, back through the DMR.                 |

**Table 2:** The three cases carried in the Load Flow tab. The first two share the same equations; only the third involves the DMR.

### 3 Model

The link is represented by three elements in series: the rectifier station, the DC line, and the inverter station. Each is a single algebraic relation.

#### 3.1 Line resistance

The resistance of one pole conductor over the whole route is the specific resistance of a single subconductor, divided by the number of subconductors in the bundle (they are electrically in parallel), converted to  $\Omega/\text{mi}$  and multiplied by the length:

$$R_{HV} = \frac{r'_{HV} \cdot 5.28}{n_{HV}} L, \quad R_{DMR} = \frac{r'_{DMR} \cdot 5.28}{n_{DMR}} L. \quad (1)$$

The resistance that matters for the flow is that of the complete circuit — out and back — which depends on the configuration:

$$R_{loop} = \begin{cases} 2 R_{HV} & \text{symmetrical monopole, and balanced bipole,} \\ R_{HV} + R_{DMR} & \text{bipole with one pole out (DMR return).} \end{cases} \quad (2)$$

Equation (1) is a DC resistance at the temperature of the underlying conductor table. No temperature correction, skin effect or proximity effect is applied; on a DC line the first is the only one that matters and it is handled by choosing the conductor data at the design temperature (Section 7).

#### 3.2 Converter stations

Station losses are taken as a fraction of the power passing through the station. The two stations therefore differ only in which side of the conversion is known:

$$\text{rectifier (AC} \rightarrow \text{DC): } P_{dc,rec} = P_{ac,rec} (1 - p_{rec}) \iff P_{ac,rec} = \frac{P_{dc,rec}}{1 - p_{rec}}, \quad (3)$$

$$\text{inverter (DC} \rightarrow \text{AC): } P_{ac,inv} = P_{dc,inv} (1 - p_{inv}) \iff P_{dc,inv} = \frac{P_{ac,inv}}{1 - p_{inv}}. \quad (4)$$

A single fraction lumps everything inside the station fence: valve conduction and switching losses, converter transformer and reactor losses, cooling and auxiliary supply. The loss fraction is a straight percentage of throughput, so it scales linearly with power and does not capture the no-load component of a real loss curve; see Section 7.

Because power is specified at the receiving AC terminals, Equation (4) is the *entry point* of the whole calculation: it converts the given  $P_{ac,inv}$  into the DC power that must arrive at the inverter DC terminals.

#### 3.3 DC line

With the DC voltage held at  $V_d$  pole-to-ground at the rectifier, the power leaving the rectifier and the resistive loss of the line are

$$P_{dc,rec} = k V_d I_d, \quad P_{loss,line} = R_{loop} I_d^2, \quad k = \begin{cases} 2 & \text{two poles energised,} \\ 1 & \text{one pole out,} \end{cases} \quad (5)$$

and the power arriving at the inverter DC terminals is the difference:

$$P_{dc,inv} = k V_d I_d - R_{loop} I_d^2. \quad (6)$$

Equation (6) is the load flow equation of the link. Everything in Section 4 follows from solving it.

The loss splits between the conductors in proportion to their resistances, which is reported separately because the DMR contribution is what distinguishes the outage case:

$$P_{loss,HV} = \begin{cases} 2 R_{HV} I_d^2 & \text{balanced,} \\ R_{HV} I_d^2 & \text{one pole out,} \end{cases} \quad P_{loss,DMR} = \begin{cases} 0 & \text{balanced,} \\ R_{DMR} I_d^2 & \text{one pole out.} \end{cases} \quad (7)$$

### 3.4 Receiving-end voltage

The voltage at the far end of the line is the sending-end voltage less the drop along the conductors the current actually traverses. With two poles energised the arrangement is symmetric — each conductor drops  $I_d R_{HV}$  and the midpoint stays at ground — so the pole-to-ground voltage at the inverter is

$$V_{d,inv} = V_d - I_d R_{HV} \quad (\text{two poles energised}), \quad (8)$$

which is consistent with Equation (6), since  $2V_{d,inv}I_d = 2V_dI_d - 2R_{HV}I_d^2$ . With one pole out the whole loop drop appears between the healthy pole and the DMR, so the corresponding figure is a pole-to-DMR voltage:

$$V_{d,inv} = V_d - I_d R_{loop} \quad (\text{one pole out, pole-to-DMR}). \quad (9)$$

## 4 Solution of the load flow

### 4.1 Why the current is not explicit

If both the power and the voltage were specified at the same end of the line, the current would follow from a division. They are not: power is fixed at the receiving end and voltage at the sending end, and the two are separated by a loss that itself depends on the current. Substituting the target power into Equation (6) gives a quadratic in  $I_d$  rather than a linear relation. It is a quadratic that can be solved in closed form, so no iteration is needed — which is what keeps the calculation a single row of the spreadsheet.

### 4.2 Two poles energised (symmetrical monopole, balanced bipole)

Start from the required DC power at the inverter, Equation (4):

$$P_{dc,inv} = \frac{P_{ac,inv}}{1 - p_{inv}}. \quad (10)$$

Insert it into Equation (6) with  $k = 2$  and collect terms:

$$R_{loop} I_d^2 - 2V_d I_d + P_{dc,inv} = 0. \quad (11)$$

The roots are

$$I_d = \frac{2V_d \pm \sqrt{4V_d^2 - 4R_{loop}P_{dc,inv}}}{2R_{loop}} = \frac{V_d \pm \sqrt{V_d^2 - R_{loop}P_{dc,inv}}}{R_{loop}},$$

and the operating point is the **lower** root:

$$I_d = \frac{V_d - \sqrt{V_d^2 - R_{loop}P_{dc,inv}}}{R_{loop}} \quad (12)$$

with  $V_d$  in kV,  $R_{loop}$  in  $\Omega$ ,  $P_{dc,inv}$  in MW and  $I_d$  in kA.

Both roots deliver the same power to the inverter, but they do it at different efficiencies: the upper root is the high-current solution on the far side of the line's maximum power point, where more current is pushed through the line only to burn the extra in  $I^2R$ . It is not an operating point any converter control would choose, and it lies far outside the current rating of the conductors. The lower root is the physical one.

### 4.3 Maximum transferable power

The square root in Equation (12) becomes imaginary when the requested power exceeds what the line can deliver at that voltage. The limit follows from  $dP_{dc,inv}/dI_d = 0$  applied to Equation (6):

$$I_d|_{P_{max}} = \frac{k V_d}{2R_{loop}}, \quad P_{max} = \frac{k^2 V_d^2}{4R_{loop}} = \begin{cases} \frac{V_d^2}{R_{loop}} & \text{two poles energised,} \\ \frac{V_d^2}{4R_{loop}} & \text{one pole out.} \end{cases} \quad (13)$$

$P_{max}$  is reported as a solvability check, not as a capability: at that point half the sending-end power is dissipated in the line. For any sensible design it sits an order of magnitude above the rating, and a real link is limited by conductor thermal rating and valve current long before it. Its practical use is diagnostic — a #NUM! in the current row means the requested power, voltage and resistance combination has no solution, and comparing against  $P_{max}$  shows immediately which input is at fault.

#### 4.4 Forward chain

With the current known, the rest of the quantities follow directly:

$$P_{dc,rec} = k V_d I_d \quad [\text{Eq. (5)}] \quad (14)$$

$$P_{loss,line} = R_{loop} I_d^2 \quad [\text{Eq. (5)}, \text{ split by Eq. (7)}] \quad (15)$$

$$V_{d,inv} = V_d - I_d R_{HV} \quad [\text{Eq. (8)}] \quad (16)$$

$$P_{ac,rec} = \frac{P_{dc,rec}}{1 - p_{rec}} \quad [\text{Eq. (3)}] \quad (17)$$

$$P_{loss,total} = P_{ac,rec} - P_{ac,inv} \quad \text{AC-to-AC loss} \quad (18)$$

$$\eta = \frac{P_{ac,inv}}{P_{ac,rec}} \quad \text{overall efficiency} \quad (19)$$

The three loss contributions add up to the total, which is the closure check built into the workbook:

$$P_{ac,rec} - P_{ac,inv} = \underbrace{p_{rec} P_{ac,rec}}_{\text{rectifier}} + \underbrace{R_{loop} I_d^2}_{\text{line}} + \underbrace{p_{inv} P_{dc,inv}}_{\text{inverter}}. \quad (20)$$

#### 4.5 Bipole with one pole out

The outage case is solved forward rather than backward, because the input is a current and not a power. Taking the pole current from the balanced bipole solution,

$$I_{d,mono} = I_{d,bipole}, \quad R_{loop} = R_{HV} + R_{DMR}, \quad (21)$$

the chain runs in the opposite direction to Equation (10):

$$P_{dc,rec} = V_d I_d \quad (22)$$

$$P_{loss,line} = R_{HV} I_d^2 + R_{DMR} I_d^2 = R_{loop} I_d^2 \quad (23)$$

$$P_{dc,inv} = P_{dc,rec} - P_{loss,line} \quad (24)$$

$$P_{ac,inv} = P_{dc,inv} (1 - p_{inv}) \quad (25)$$

so that the transferable power is a *result* of the case rather than an input. The residual capability is reported as a fraction of the bipolar rating,

$$\frac{P_{ac,inv,mono}}{P_{ac,inv,bipole}}, \quad (26)$$

which lands slightly below 50 % — the pole current is unchanged but only one pole carries power, and the DMR return is more resistive than the HV bundle it replaces.

#### 4.6 Current margin

The margin does not enter the flow. It scales the operating current to a rating figure used for conductor selection, valve and bushing ratings, switchgear and protection settings:

$$I_{rated} = (1 + m) I_d. \quad (27)$$

Losses, voltage drops and powers are all reported at the operating current  $I_d$ , not at  $I_{rated}$ ; the margined value is a sizing quantity only. If a worst-case loss figure is wanted, the calculation is re-run with the power input raised rather than by substituting  $I_{rated}$  into the loss equation, which would be inconsistent with the stated power.

## 5 Implementation in the workbook

### 5.1 Layout

The calculation lives on the **Load Flow** tab of the HVDC calculation workbook. Column A carries the quantity, column B its unit, and the three result columns are the three cases of Table 2: column C the symmetrical monopole, column D the balanced bipole, column E the bipole with one pole out. Column E takes its plant data from column D by reference, so the outage case cannot drift out of step with the bipole it belongs to; only the flow rows differ. Input cells are shaded, results are shaded in the second house colour, matching the convention already used on the Power and Losses tabs.

### 5.2 Inputs

| Row   | Input                                       | Symbol              | Remark                                                       |
|-------|---------------------------------------------|---------------------|--------------------------------------------------------------|
| 6     | AC power at receiving end                   | $P_{ac,inv}$        | At the inverter AC terminals; the design target.             |
| 7     | DC voltage pole-to-ground                   | $V_d$               | At the rectifier DC terminals.                               |
| 8     | DC line length                              | $L$                 | Route length, miles.                                         |
| 9–11  | HV conductor, bundles, specific resistance  | $n_{HV}, r'_{HV}$   | Per subconductor, $\Omega/1000$ ft.                          |
| 12–14 | DMR conductor, bundles, specific resistance | $n_{DMR}, r'_{DMR}$ | Separate input: the DMR is generally not the pole conductor. |
| 15    | Losses inverter                             | $p_{inv}$           | Fraction of station throughput.                              |
| 16    | Losses rectifier                            | $p_{rec}$           | Fraction of station throughput.                              |
| 17    | Margin for DC current                       | $m$                 | Applied to the rated current only.                           |

Table 3: Input rows of the Load Flow tab.

The DC voltage is an input the original request did not list, but the flow cannot be determined without it: the same power at the same length gives a different current, loss and voltage drop at every voltage level, and it is the voltage that fixes which of them applies. It is entered pole-to-ground, the same convention as the Losses tab.

### 5.3 Formulas as entered

Table 4 maps every calculated row to the equation it implements. Cell references are for column C (symmetrical monopole); columns D and E are the same relations on their own column, with the two differences noted at the foot of the table.

Two rows differ in the outage column, and they are the whole of the difference between a balanced and an unbalanced bipole:

- **Row 22**, loop resistance, becomes =E20+E21 — one pole conductor plus the DMR, Equation (2).
- **Row 26**, current, becomes =D26 — the case is defined by holding the pole current, so rows 25 and 28 reverse direction: row 28 is =E7\*E26/1000 (single pole,  $k = 1$ ) and row 25 becomes =E28-E31, the forward chain of Equations (22)–(25).

### 5.4 Row 33 as a closure check

In columns C and D, row 33 recomputes  $P_{ac,inv}$  from the DC power that row 25 derived from it. It therefore has to reproduce the input in row 6 exactly:

$$\frac{P_{ac,inv}}{1 - p_{inv}} (1 - p_{inv}) = P_{ac,inv}.$$

| Row | Formula as entered                | Quantity         | Eq.  |
|-----|-----------------------------------|------------------|------|
| 20  | =C11*5.28/C10*C8                  | $R_{HV}$         | (1)  |
| 21  | =D14*5.28/D13*D8                  | $R_{DMR}$        | (1)  |
| 22  | =2*C20                            | $R_{loop}$       | (2)  |
| 25  | =C6/(1-C15)                       | $P_{dc,inv}$     | (10) |
| 26  | =(C7-SQRT(C7^2-C22*C25))/C22*1000 | $I_d$            | (12) |
| 27  | =C26*(1+C17)                      | $I_{rated}$      | (27) |
| 28  | =2*C7*C26/1000                    | $P_{dc,rec}$     | (14) |
| 29  | =C22*(C26/1000)^2                 | $P_{loss,HV}$    | (7)  |
| 30  | 0                                 | $P_{loss,DMR}$   | (7)  |
| 31  | =C29+C30                          | $P_{loss,line}$  | (15) |
| 32  | =C7-C26/1000*C20                  | $V_{d,inv}$      | (8)  |
| 33  | =C25*(1-C15)                      | $P_{ac,inv}$     | (4)  |
| 34  | =C28/(1-C16)                      | $P_{ac,rec}$     | (17) |
| 35  | =C34-C33                          | $P_{loss,total}$ | (18) |
| 36  | =C33/C34*100                      | $\eta$           | (19) |
| 39  | =C7^2/C22                         | $P_{max}$        | (13) |

**Table 4:** Calculated rows of the Load Flow tab and the equations they implement. The 1000 factors convert between the kiloamperes of the equations and the amperes reported in row 26.

Any discrepancy between rows 6 and 33 means a formula in the chain has been disturbed — it is a cheap, always-visible check on the sheet. In column E, row 33 is not a check but the answer: it is the power the link can still deliver on one pole.

## 6 Reference case

### 6.1 Inputs

The reference case uses the project data already carried on the Losses tab: a 200 mi line on Bluebird ACSR, two subconductors per pole, with the DMR provisionally taken as a single Bluebird conductor. Station losses are 0.8 % per station and the current margin 10 %.

| Input                        | Unit             | Sym. monopole | Bipole  |
|------------------------------|------------------|---------------|---------|
| AC power at receiving end    | MW               | 1,500         | 3,000   |
| DC voltage pole-to-ground    | kV               | 400           | 525     |
| DC line length               | mi               | 200           | 200     |
| HV subconductors per pole    | —                | 2             | 2       |
| Specific HV resistance       | $\Omega/1000$ ft | 0.00801       | 0.00801 |
| DMR subconductors            | —                | n/a           | 1       |
| Specific DMR resistance      | $\Omega/1000$ ft | n/a           | 0.00801 |
| Losses per converter station | %                | 0.8           | 0.8     |
| Margin for DC current        | %                | 10            | 10      |

Table 5: Reference case inputs.

### 6.2 Line resistance

From Equation (1), identical for both configurations because they share the conductor and the length:

$$R_{HV} = \frac{0.00801 \times 5.28}{2} \times 200 = 4.229 \Omega, \quad R_{DMR} = \frac{0.00801 \times 5.28}{1} \times 200 = 8.459 \Omega,$$

so  $R_{loop} = 8.459 \Omega$  with two poles energised and  $12.688 \Omega$  with the return through the DMR.

### 6.3 Worked step-through, bipole

Taking the bipole column and following Section 4:

$$P_{dc,inv} = \frac{3000}{1 - 0.008} = 3024.2 \text{ MW} \quad \text{Eq. (10)}$$

$$I_d = \frac{525 - \sqrt{525^2 - 8.459 \times 3024.2}}{8.459} = \frac{525 - 500.04}{8.459} = 2.950 \text{ kA} \quad \text{Eq. (12)}$$

$$P_{dc,rec} = 2 \times 525 \times 2.950 = 3097.8 \text{ MW} \quad \text{Eq. (14)}$$

$$P_{loss,line} = 8.459 \times 2.950^2 = 73.6 \text{ MW} \quad \text{Eq. (15)}$$

$$V_{d,inv} = 525 - 2.950 \times 4.229 = 512.5 \text{ kV} \quad \text{Eq. (8)}$$

$$P_{ac,rec} = \frac{3097.8}{1 - 0.008} = 3122.8 \text{ MW} \quad \text{Eq. (17)}$$

$$\eta = \frac{3000}{3122.8} = 96.07 \% \quad \text{Eq. (19)}$$

The closure check of Equation (20) balances:  $24.98 + 73.63 + 24.19 = 122.8 \text{ MW} = P_{ac,rec} - P_{ac,inv}$ .

### 6.4 Results

### 6.5 Reading the results

Three points are worth drawing out.

| Result                          | Unit     | Sym. monopole | Bipole  | Bipole, 1 pole out |
|---------------------------------|----------|---------------|---------|--------------------|
| Loop resistance                 | $\Omega$ | 8.459         | 8.459   | 12.688             |
| DC power at inverter terminals  | MW       | 1,512.1       | 3,024.2 | 1,438.5            |
| DC current, operating           | A        | 1,929         | 2,950   | 2,950              |
| DC current, rated (10 % margin) | A        | 2,122         | 3,245   | 3,245              |
| DC power at rectifier terminals | MW       | 1,543.6       | 3,097.8 | 1,548.9            |
| Line losses, HV conductors      | MW       | 31.5          | 73.6    | 36.8               |
| Line losses, DMR                | MW       | —             | 0.0     | 73.6               |
| Total DC line losses            | MW       | 31.5          | 73.6    | 110.4              |
| DC voltage at inverter          | kV       | 391.8         | 512.5   | 487.6              |
| AC power at receiving end       | MW       | 1,500.0       | 3,000.0 | 1,427.0            |
| AC power at sending end         | MW       | 1,556.0       | 3,122.8 | 1,561.4            |
| Total losses, AC to AC          | MW       | 56.0          | 122.8   | 134.4              |
| Overall efficiency              | %        | 96.40         | 96.07   | 91.39              |
| Maximum transferable power      | MW       | 18,916        | 32,585  | 5,431              |
| Power vs. bipole rating         | %        | —             | 100.0   | 47.6               |

**Table 6:** Reference case results. The one-pole-out column runs at the same pole current as the balanced bipole.

**The DMR dominates the outage case.** At 110.4 MW the line loss in the one-pole-out condition is half again the loss of the balanced bipole at twice the power, and two thirds of it is in the DMR. That is a direct consequence of the placeholder assumption of a single Bluebird conductor: the return path is twice the resistance of the two-conductor bundle it stands in for. Sizing the DMR closer to the pole bundle moves the residual capability up toward 50 % and cuts the outage loss substantially — this is the first input to revisit with real conductor data.

**Defining power at the receiving end raises the current.** The Losses tab derives the bipole current directly as  $P/(2V_d) = 2857$  A; the load flow gives 2950 A, some 3 % higher. The difference is not an inconsistency but a difference in what the power figure means: the Losses tab takes 3000 MW as the power *leaving* the rectifier, whereas here 3000 MW must *arrive* at the receiving AC terminals, so the current has to cover the line and converter losses as well. The load flow figure is the one to size equipment against when the guarantee is written at the point of delivery.

**The line, not the converters, sets the loss.** At 0.8 % per station the two converters together account for 49 MW of the 123 MW lost in the bipole case, against 74 MW in the conductors. Line loss scales with the square of the current and therefore inversely with the square of the voltage; converter loss scales linearly with power. That asymmetry is the mechanism behind the voltage-level trade study, and this tab is the tool for it — the comparison between the 400 kV monopole and the 525 kV bipole columns is a first instance of exactly that.

## 7 Assumptions, limitations and open items

### 7.1 Assumptions

| Assumption                                | Consequence and justification                                                                                                                                                                                                                                  |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purely resistive DC line                  | No inductance or capacitance appears in the equations. Correct for a steady-state DC flow; line charging and inductance matter only for transients and fault studies.                                                                                          |
| Constant conductor resistance             | The specific resistance is taken at the temperature of the conductor table. Resistance rises roughly 0.4 %/K for aluminium, so a hot-day loss figure is a few percent higher than reported here. Handled by choosing conductor data at the design temperature. |
| Ideal voltage regulation at the rectifier | $V_d$ holds exactly at the sending terminals irrespective of loading. Real control holds it within a droop band.                                                                                                                                               |
| Station losses proportional to throughput | No no-load term, so light-load efficiency is overstated and full-load efficiency is representative. Adequate for screening; a two-term model ( $p_0 + p_1 P$ ) is the natural refinement.                                                                      |
| Balanced bipole carries no DMR current    | Pole currents are taken as exactly equal. In service a small unbalance flows in the DMR; its loss contribution is second order (the square of a few percent of $I_d$ ).                                                                                        |
| One pole out at unchanged pole current    | Keeps the outage case inside the ratings established for normal operation. An alternative case at unchanged <i>power</i> would double the pole current and require its own rating basis.                                                                       |
| Single operating point                    | The tab evaluates one dispatch, not a duty cycle. Annual energy losses require a loading profile and a separate loss-energy calculation.                                                                                                                       |

Table 7: Assumptions underlying the load flow calculation.

### 7.2 Outside the scope

The calculation deliberately stops short of the following, each of which belongs to a study rather than to a spreadsheet cell:

- **Reactive power and AC-side voltage.** Only active power crosses the converter boundary in this model. Converter reactive capability and AC network voltage support are covered by the AC calculations on the Power tab and, ultimately, by interconnection studies.
- **Ground and electrode return.** Only metallic return is modelled. A bipole operating with earth return has a different loop resistance and a set of electrode, corrosion and interference constraints that sit outside a load flow.
- **Thermal rating of the conductor.** The tab reports the current that flows; whether the conductor can carry it continuously at the ambient and wind conditions of the route is an ampacity calculation.
- **Converter overload and short-term ratings.** The margin of Equation (27) is a continuous rating margin, not an overload capability.
- **Multi-terminal and intermediate taps.** The equations assume a two-terminal, point-to-point link.
- **Dynamics.** Control response, ramp rates, pole-block transients and the transition into the one-pole-out condition are all outside a steady-state flow.

### 7.3 Open items

- (1) **DMR conductor selection.** The reference case uses a single Bluebird conductor as a placeholder and the outage-case loss is sensitive to it (Section 6). To be replaced with the selected DMR conductor and bundle.

- (2) **Unit label on the Losses tab.** The specific resistance row is labelled  $\Omega/\text{mi}$  but holds  $\Omega/1000 \text{ ft}$ ; the 5.28 factor in the loss formula confirms it. The label should be corrected so the two tabs read consistently.
- (3) **Converter loss figure.** 0.8 % per station is a generic VSC-class value. To be replaced with vendor loss curves, at which point the two-term loss model of Table 7 becomes worthwhile.
- (4) **Temperature basis for conductor resistance.** To be stated explicitly alongside the conductor data, so the reported losses carry a defined temperature.
- (5) **Loss-energy evaluation.** A loading profile and an annual loss-energy figure are the natural extension, since it is energy rather than peak loss that enters an economic comparison of voltage levels and conductor sizes.